

He Min

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EDUCATIONAL BACKGROUND

Nanyang Technological University

Master of Science in Signal Processing and Machine Learning

Chongqing University of Technology

Bachelor of Science in Computer Science and Technology (Honours)

• **GPA:** 3.96 / 5.0 (Top 1%)

Singapore

2026.01-2027.07

Chongqing, China

2021.09-2025.06

INTERNSHIP EXPERIENCE

Lenovo (Shanghai) Information Technology Co., Ltd. (LR Robotics Algorithm)

2026.05-2026.08

Research Intern in Embodied AI for Robotics

Project Overview: Leveraging the Lenovo Morningstar robot as the platform, a collaborative intelligent agent system has been constructed, comprising the task-planning "mastermind" NavAgent and the dual-system navigator InternVLA-N1-DualVLN. NavAgent utilizes the LangGraph state machine and MCP to orchestrate the ROS2 toolkit, handling global task parsing, memory storage, telemetry self-checks, and anomaly diagnosis. DualVLN manages specific motion generation: System 2 (2 Hz; fine-tuned QwenVL-2.5) predicts 2D pixel sub-goals, while System 1 (30 Hz; NextDiT) outputs high-frequency physical trajectories. SR increased by 10.1%, SPL increased by 6.7%, NE decreased by 17.6%, and StR decreased by 32.5%.

PROJECT EXPERIENCE

Fine-tuning and Closed-loop Evaluation of SmolVLA Robot Fruit-Grasping Strategies Based on Multimodal Synthetic Data

2026.07- Present

Project Overview: This project establishes an end-to-end development pipeline on AMD GPUs (ROCm) comprising multi-objective synthetic data generation, VLA policy fine-tuning, and closed-loop evaluation. Using the Genesis IK expert controller, the project captures grasping trajectories of a Franka robotic arm for fruits of various geometries and aggregates them into a LeRobot-standard multimodal dataset. Based on this, SmolVLA is fine-tuned, and closed-loop placement validation is conducted in scenarios involving multi-objective random perturbations.

OV-LidarMap: Open-Set Instance-Level 3D Semantic Mapping with Rotating LiDAR and Panoramic Vision

Co-first Author, **IEEE RA-L (SCI Q2 TOP, Under Review)** [[arXiv](#)] Supervisor: Prof. Lihua Xie 2026.01-2026.03

Project Overview: To address the issues of lacking spatial reasoning capabilities in Large Language Models (LLMs) and the "context explosion" problem within large-scale 3D environments, an omnidirectional perception-driven embodied AI agent framework was developed using Python, PyTorch, and ROS 2. By constructing a dynamic scene graph as the agent's world model and employing a multi-resolution spatial attention mechanism, the framework achieves zero-shot long-range navigation with a Success Rate (SR) of 93.18% and a 69.98% reduction in token costs.

PUBLICATION

He, M. (2025). A Review of Data Fusion and Deep Learning Models for Multimodal Sentiment Analysis. In Proceedings of E-commerce and Artificial Intelligence (ECAI 2024). [[SciTePress](#)]

AWARDS & HONORS

National Scholarship (2024); First Prize, National College Student Mathematics Competition (2023); Third Prize, National Finals of the Team Programming Ladder Tournament (2023)

TECHNOLOGY STACK

• **Embodied Intelligence & Large Model Agent:** Proficient in **VLA (Vision-Language-Action)** end-to-end policy fine-tuning and imitation learning; skilled in **LangGraph** state machine design and encapsulation of **MCP** Protocol tool ecosystems; familiar with **CoT** reasoning chains and large model adaptive resolution / prompt compression technologies.

• **Robotic Systems, Perception and Simulation:** Proficient in **ROS 2** communication mechanisms; familiar with physics simulation engines such as **Genesis** and the **LeRobot** embodied dataset standard; experienced in **SAM2** object segmentation, **Open3D** point cloud processing and **KDTree** search; with hands-on experience in constructing 3D dynamic scene graphs.

• **Systems Engineering & Concurrent Programming:** Proficient in **Python** and **C++**; familiar with **Linux**, **Docker** containerization, and **Git** version control; experienced in encapsulating **low-level control logic** for **multi-threaded applications** and conducting **hardware-software integration and debugging**.